Middle Primary Division

Questions 1 to 10, 3 marks each
1-10 题，每题 3 分

1. What is the value of the digit 7 in the number 57021?

(A) 7 (B) 70 (C) 700 (D) 7000 (E) 70 000

The digit 7 is in the fourth place value from the right, which is the thousands place. Therefore, the value of the digit 7 is 7000,

hence (D).

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2. Which one of these numbers is closest to 365?

(A) 265 (B) 356 (C) 361 (D) 368 (E) 375

Since the given numbers are in order, we only need to look at 361, which is 4 less than 365, and 368, which is 3 more than 365. So 368 is closest,

hence (D).

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3. Which pattern has exactly 8 dots?

(A) (B) (C) (D) (E)

Only the first pattern (A) has 8 dots,

hence (A).

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4. Marlie was born in 2016. In which year will she turn 14?

(A) 2026 (B) 2028 (C) 2030 (D) 2032 (E) 2034

Since $2016 + 14 = 2030$, she will turn 14 in 2030,

hence (C).

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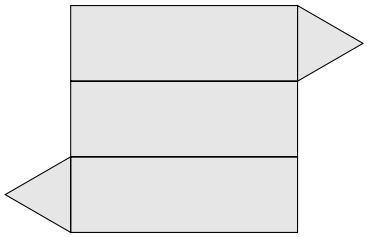
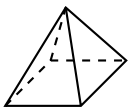
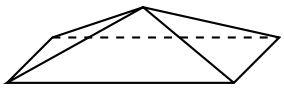
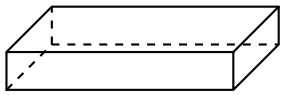
5. Karl cuts out this cardboard shape then folds it along the lines to make a box.

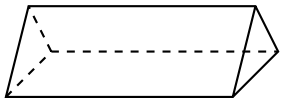
What does the box look like?

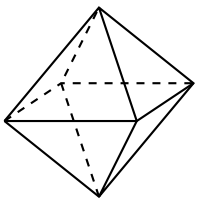


(A) 

(B) 

(C) 

(D) 

(E) 

The three rectangles in the net must wrap around to form the three sides of a triangular prism, and then the two triangles are the ends,

hence (D).

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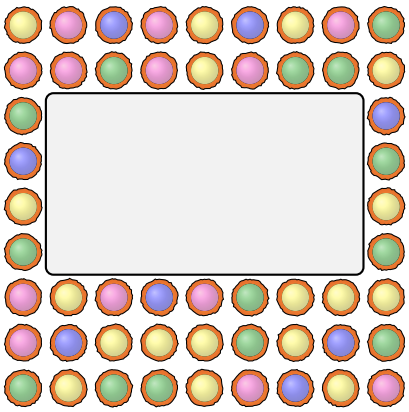
6. Sam has arranged some cupcakes in equal rows on her kitchen table.

To work out how to pack them for the party, she places a box over a number of them, as shown.

How many are covered by the box?



(A) 24 (B) 28 (C) 32 (D) 45 (E) 81

Counting cupcakes along each side, the covered cupcakes are 4 rows of 7, so $4 \times 7 = 28$ cupcakes are covered,

hence (B).

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7. In the word MATHS, which letter has the most lines of symmetry?

(A) A (B) H (C) M (D) S (E) T

The letter S has zero lines of symmetry. The letters M, A & T all have one vertical line of symmetry. The letter H has two lines of symmetry, horizontal and vertical,

hence (B).

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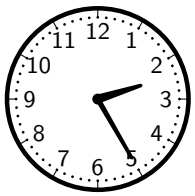
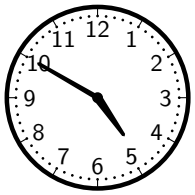
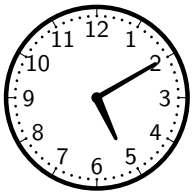
8. Which one of these clock faces shows 10 minutes past 5?

(A)

(B) 

(C)

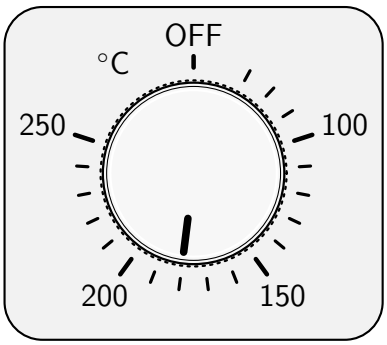
(D) 

(E) 

To have 10 minutes past the hour, the minutes hand should be on the digit 2. The hours hand will be slightly past 5,

hence (E).

9. Evan set the temperature dial on his oven as shown. What temperature is it set to?



(A) 155 °C (B) 170 °C

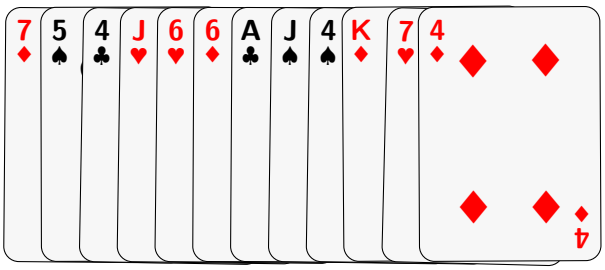
(C) 180 °C (D) 220 °C

(E) 230 °C

Between 150 and 200 there are 5 intervals, so each interval represents 10 °C. The dial is set 3 intervals above 150 °C which is therefore 180 °C,

hence (C).

10. If I choose one of these cards at random, which of the following am I least likely to get?



(A) A four

(B) A seven

(C) A spade ♠

(D) A heart ♥

(E) A diamond ♦

For each of spades, diamonds, hearts and fours, there are 3 cards. However, there are only 2 sevens. So I am least likely to get a seven,

hence (B).

Questions 11 to 20, 4 marks each
11-20 题，每题 4 分

11. In this puzzle, every row and every column must contain the numbers 1, 3 and 5, once each.

If I solve the puzzle then multiply the numbers in the two shaded boxes, what is the answer?

3		
	5	

(A) 1 (B) 3 (C) 5 (D) 15 (E) 25

The first column can't be $3, 5, 1$ so it must be $3, 1, 5$.

Similarly, the second row must be $1, 5, 3$.

The lower right cell can't be 3 or 5, so must be 1.

So both shaded cells are 1, and $1 \times 1 = 1$,

3	1	5
1	5	3
5	3	1

hence (A).

12. Gabe has this birthday chart on his classroom wall. It shows how many people in the class were born in each season.

The number for Summer has been rubbed off but Gabe knows that more people were born in Summer than in any other season.

What is the smallest possible number of people in the class?

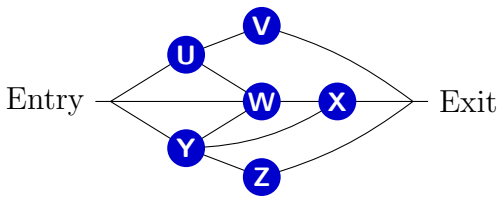
Summer	
Autumn	5
Winter	8
Spring	7

(A) 20 (B) 21 (C) 25 (D) 28 (E) 29

If more people were born in Summer than in any other season, then at least 9 people were born in Summer. This means that the smallest number of people in the class is $9 + 5 + 8 + 7 = 29$,

hence (E).

13. Nullah's local park has six fountains and many paths, as this map shows. His run through the park always follows the paths that lead forward and he always passes exactly two fountains.



How many different ways are there for Nullah to pass exactly two fountains in the park?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

If Nullah first runs to fountain U, then only fountain V can be next.

If Nullah first runs to W, then only X can be next.

If Nullah first runs to Y, then either X or Z can be next.

These are the only 4 possibilities,

hence (C).

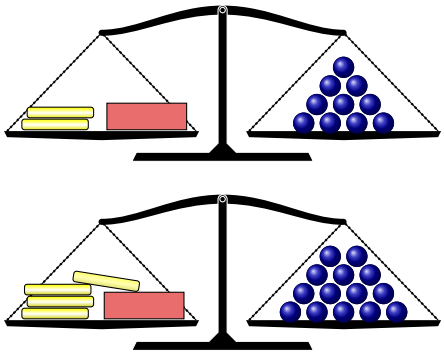
14. Justine has some balls, rods and bricks.

Two rods and one brick weigh the same as 10 balls.

Four rods and one brick weigh the same as 14 balls.

How many balls weigh the same as one brick?

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7



Alternative 1

Two extra rods requires 4 balls to balance, so each rod weighs the same as 2 balls.

Then two rods and one brick weigh the same as 4 balls and one brick. This also weighs the same as 10 balls.

So one brick weighs the same as 6 balls,

hence (D).

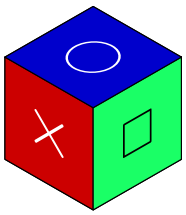
Alternative 2

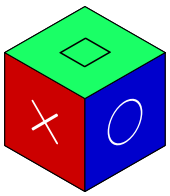
Doubling the first amount, 4 rods plus 2 bricks weigh the same as 20 balls. This is 1 brick more, and 6 balls more than the second amount. So 1 brick weighs the same as 6 balls,

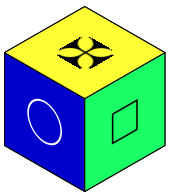
hence (D).

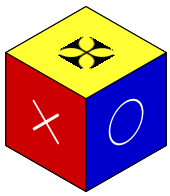
15. The cube pictured has six different faces.

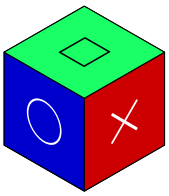
Which of the following is **not** a possible picture of this cube?

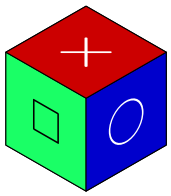


(A) 

(B) 

(C) 

(D) 

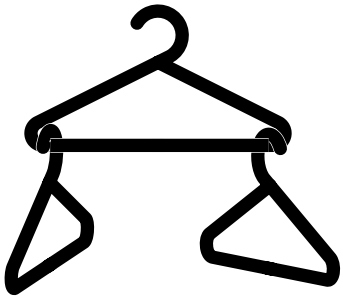
(E) 

Pictures (D) and (E) are possible by rotating the diagram of the cube by 120° anticlockwise and clockwise, respectively. Picture (A) shows the same faces, but the order of the red-blue-green faces is clockwise in the question and anticlockwise in (A). So (A) is not possible. The other two options are possible: (B) by having the red face downwards and (C) by having the green face downwards. Consequently the only picture that is not possible is (A),

hence (A).

16. Coco is making a hanging mobile from coathangers.

Starting with 1 coathanger in the first layer, she hangs 2 coathangers in the second layer, 1 at each end of the first, as shown.



She works along each layer following the same pattern, so that every coathanger has 2 new coathangers hanging from it.

Coco stops when she has finished 6 layers. How many coathangers are in her mobile?

(A) Between 20 and 30 (B) Between 30 and 40 (C) Between 40 and 50 (D) Between 50 and 60 (E) Between 60 and 70

Since she hangs 2 coat hangers to the ends of each one in the layer above, the total number of coat hangers in each layer doubles at each stage. So in the first 6 layers the numbers of coat hangers are

$$1,\ 2,\ 4,\ 8,\ 16,\ 32$$

and adding these together gives 63 in all,

hence (E).

17. A shop has a special deal where you can buy two different items and get the cheaper item at half price. Roberta bought two of these items, and paid \$16. Which two items did she buy?

(A) A birthday card and a magazine

(B) A gift card and a birthday card

(C) A book and a roll of wrapping paper

(D) A gift card and a roll of wrapping paper

(E) A magazine and a roll of wrapping paper

Since Roberta paid \$16, she spent more than \$8 on the most expensive item, which is either a book or a gift card. If her expensive item was a \$10 book, then she spent \$6 on her other item at half price. But then her cheaper item would have cost \$12, which does not work. So her expensive item can't have been a book.

If her expensive item was a \$12 gift card, she spent \$4 on the other item, which must have been priced at \$8, which is the wrapping paper. So Roberta bought a gift card and a roll of wrapping paper,

hence (D).

18. Lubna likes oranges and apples, but not bananas. Bhavna likes apples and bananas, but not oranges. Preeti likes oranges and bananas, but not apples.

Their father has 6 oranges, 4 apples and 2 bananas and gives each girl 4 pieces of fruit that she likes. What is the **smallest** number of oranges that he can give to Lubna?

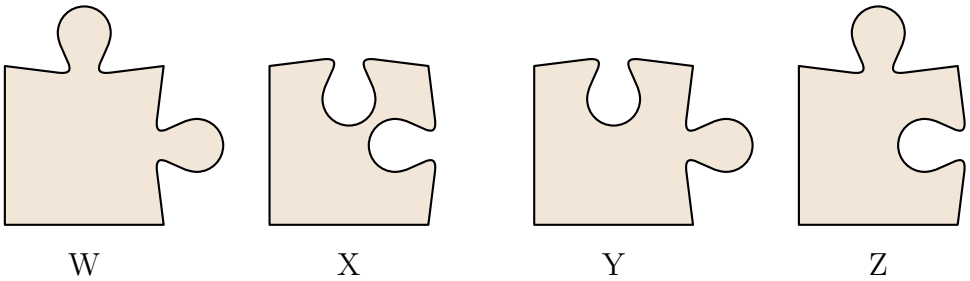
(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Since Bhavna dislikes oranges, and Preeti receives at most 4 oranges, Lubna must receive at least 2 oranges.

If Lubna receives exactly 2 oranges, then she also receives 2 apples. Also Preeti receives 4 oranges and finally Bhavna receives 2 apples and 2 bananas. This complete solution confirms that the minimum number of oranges that Lubna receives is 2,

hence (C).

19. A simple jigsaw puzzle has four pieces labelled W, X, Y and Z, shown here face up.

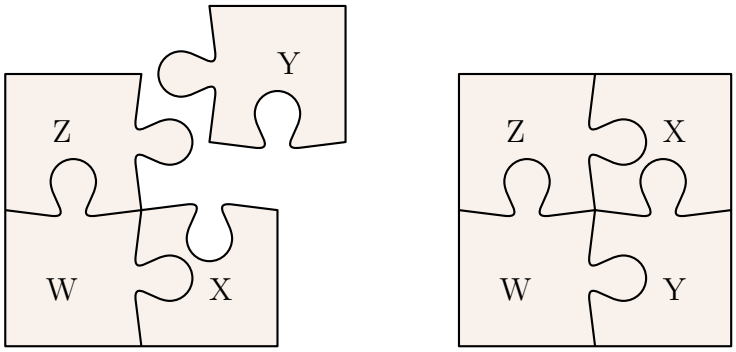


When the puzzle is completed, which of the following pieces are **not** next to each other?

(A) W and Y (B) W and Z (C) X and Y

(D) X and Z (E) Y and Z

Start with W in the bottom-left corner and consider the piece to its right. To make the straight edge along the bottom, this piece can only be X or Y (either one rotated anticlockwise by 90°). If it is X, then Z must be placed above the W as shown in the first diagram. But then Y cannot fit in the top-right corner (another Z piece would be needed to complete the puzzle).



So the piece to the right of W must be Y and the second diagram shows how the puzzle can be completed. Of the pairs listed, only Z and Y are in opposite corners,

hence (E).

20. Minnie and Max each have a set of 4 cards numbered 1 to 4. Minnie arranges her cards into two 2-digit numbers and adds the 2-digit numbers to get a total. She chooses her numbers to make her total as small as possible.

Max does the same, except that his total is as large as possible.

What is the difference between Minnie's total and Max's total?

(A) 18 (B) 29 (C) 32 (D) 36 (E) 55

Minnie makes her numbers smallest by putting the smaller digits in the 10s position and the larger digits in the ones position. She will have either $13 + 24 = 37$ or $14 + 23 = 37$.

Max makes his numbers largest by putting the larger digits in the 10s position and the smaller digits in the ones position, giving either $31 + 42 = 73$ or $32 + 41 = 73$.

The difference between their two totals is $73 - 37 = 36$,

hence (D).

Questions 21 to 25, 5 marks each
21-25 题，每题 5 分

21. A box of chocolates weighs 416 g. The box contains 13 identical chocolates. After 4 chocolates are eaten, the box weighs 332 g.

What is the weight of the empty box?

(A) 128 g (B) 143 g (C) 204 g (D) 248 g (E) 288 g

The weight of four chocolates is $416 - 332 = 84$. Therefore one chocolate weighs $84 \div 4 = 21$ g. All 13 chocolates weigh $21 \times 13 = 273$ g. The box weighs $416 - 273 = 143$ g,

hence (B).

22. Troy makes a list of 6 numbers by writing down a starting number then writing the next 5 counting numbers, going up in steps of 1. When he adds them, the total is less than 100. He increases each number by 1, and the new total is greater than 100.

What is this new total?

(A) 101 (B) 102 (C) 103 (D) 104 (E) 105

Consider the smallest case: $1+2+3+4+5+6=21$. The next case will be $2+3+4+5+6+7=27$ since all 6 numbers have been increased by 1. The next will be $27+6=33$, then $33+6=39$ and so on. So when you add six consecutive numbers together, you will always get 21 plus a multiple of 6.

Nearest to 100 are $21+78=99$ and $21+84=105$, which can be found to be $14+15+16+17+18+19=99$ and $15+16+17+18+19+20=105$,

hence (E).

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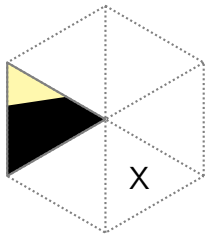
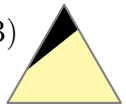
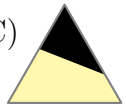
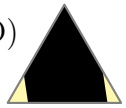
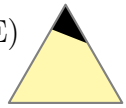
23. A regular hexagon made from yellow cardboard has a thick black stripe painted across it. The hexagon is cut up into six equilateral triangles and the pieces are mixed up. This diagram shows one of the pieces in its original position.



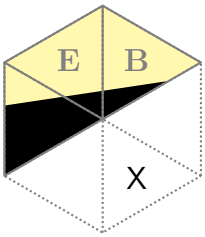
The other five pieces are shown in options (A) to (E).

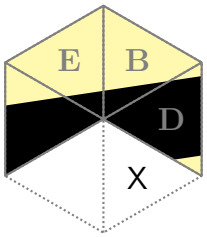
Which piece was in the position marked with the X?

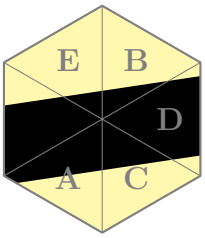
(A) (B)  (C)  (D)  (E) 

Alternative 1

Continuing the top edge of the black stripe to the right as shown in the first diagram, the top two pieces must be (E) and (B). Only piece (D) has a small yellow corner, which matches the top of the stripe as shown in the second diagram. The third diagram shows where the remaining pieces fit.







,

hence (C).

Alternative 2

The lower-left triangle has a fully black edge, so must be (A). The lower-right triangle has an edge that is slightly less than half black, which can only be (C). So triangle (C) goes in position X,

hence (C).

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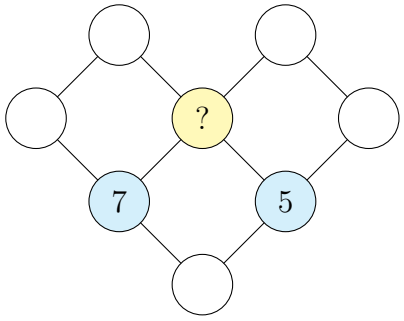
24. In this puzzle, the numbers from 1 to 8 are to be placed in the 8 circles. For each of the 3 squares, the sum of the 4 numbers on the corners must be 20.

With 5 and 7 placed as in the diagram, what number goes in the middle circle?



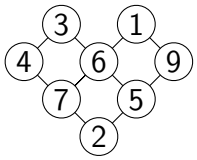
(A) 1 (B) 2 (C) 3 (D) 4 (E) 6

Alternative 1

In the bottom square, we know $5 + 7 = 12$ so the other two numbers add to $20 - 12 = 8$. Since 5 and 7 are already used, these two numbers must be 2 and 6, and there are two cases — the middle number is either 2 or 6.

If the middle number was 2, then the right square has $2 + 5 = 7$ so the other two numbers add to $20 - 7 = 13$. However, 13 can only be made as $5 + 8$ or $6 + 7$ and 5 and 7 are already used. So the middle number can't be 2.

If the middle number is 6, the bottom number is 2, and we can place 3 and 4 in the top left and 1 and 9 in the top right. This is the only solution, so the middle number must be 6. One possibility for the complete puzzle is shown,



hence (E).

Alternative 2

Try placing 1 in the middle, so that $2 + 3 + 4 + 6 + 8 = 23$ is unused. However, we have targets of 12 on the left, 14 on the right and 7 at the bottom, for a total of 33.

With 2 in the middle, $1 + 3 + 4 + 6 + 8 = 22$ is unused and the targets are $11 + 13 + 6 = 30$.

Every extra 1 in the middle decreases the unused total by 1 and decreases the total target by 3. This narrows the gap by 2. They are both equal to 18 when the middle number is 6. With 6 in the middle, the rest of the grid is easily filled in, as in the earlier solution,

hence (E).

Alternative 3

The 8 numbers add to 36, but the 3 squares add to 60. The extra 24 is from counting the numbers 5 and 7 a second time and from counting the middle number 3 times. So twice the middle number is $24 - 5 - 7 = 12$ and the middle number is 6,

hence (E).

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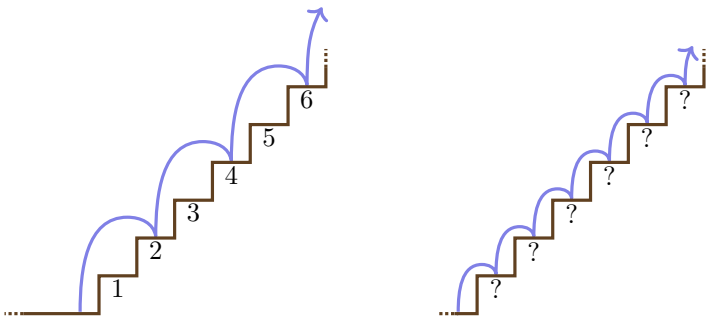
25. The stairs in Bernard's building are numbered 1, 2, 3, ... When he climbs them, he always starts by taking them two at a time, as shown below on the left. After doing this 10 times, he is tired and he takes the next 10 stairs one at a time. Then he goes back to 10 double steps, then 10 single steps, and so on.



Which of the following numbers will Bernard **not** step on?

(A) 27 (B) 51 (C) 66 (D) 79 (E) 87

The table shows the numbers up to 90 that Bernard steps on. The 'evens' type is when he has enough energy to take them two at a time. The 'all' type is when he is tired and takes them one at a time. There are 10 numbers in the list in each row.

Type	Numbers
evens	$2, 4, \ldots, 20$
all	$21, 22, \ldots, 30$
evens	$32, 34, \ldots, 50$
all	$51, 52, \ldots, 60$
evens	$62, 64, \ldots, 80$
all	$81, 82, \ldots, 90$

Numbers 27, 51, 66 and 87 are accounted for in the 2nd, 4th, 5th and 6th rows respectively. But 79 cannot be in the 5th row because it is not an even number,

hence (D).

For questions 26 to 30, shade the answer as a whole number from 0 to 999 in the space provided on the answer sheet.

问题 26-30 的答案为 0-999 之间的整数，请将答案填涂在答题卡对应区域。

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

第 26-30 题分别为 6, 7, 8, 9, 10 分。

26. Alex, Brady, Casey and Drew all share the same birthday, but are different ages. Their dad makes them a cake each, and on each cake puts birthday candles matching their ages.

- Alex's cake has 6 times as many candles as Brady's cake.
- Alex's cake has 3 times as many candles as Casey's cake.
- Alex's cake has 4 times as many candles as Drew's cake.

After the cakes are eaten, the candles are gathered up and counted. What is the smallest the total number of candles could be?

Alex's age must be a multiple of 6, a multiple of 3 and a multiple of 4. The smallest Alex's age could be is $12 = \underline{6} \times 2 = \underline{3} \times 4 = \underline{4} \times 3$. Then Brady is 2, Casey is 4 and Drew is 3.

The total number of candles is $12 + 2 + 4 + 3 = 21$,

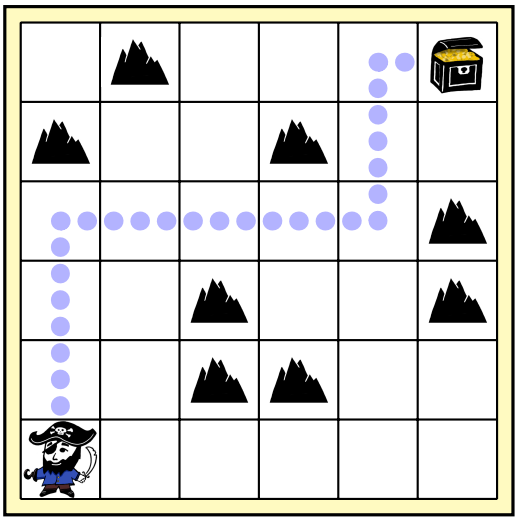
hence (21).

27. The Dread Pirate Alistair uses this map to find some buried treasure. On the map, the symbol ▲ shows dangerous territory that he must avoid.

Each move Alistair makes is either 1 square up or 1 square to the right.

One of the ways to the treasure is shown.

In how many different ways can he get to the treasure?

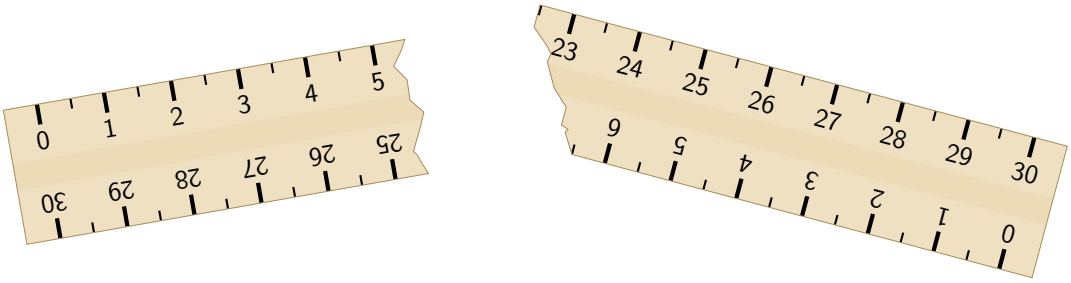


Beginning at the Dread Pirate Alistair's starting location, add in the number of possible ways he could travel to each square, given he only moves up or to the right. Note that the number of ways to get to a given square is equal to the sum of the number of ways to get to the square to its left and the number of ways to get to the square directly underneath it.

0	▲	8	8	13	$\boxed{18}$
▲	4	8	▲	5	5
1	4	4	4	5	▲
1	3	▲	0	1	▲
1	2	▲	▲	1	2
$\boxed{1}$	1	1	1	1	1

Therefore, there are $13 + 5 = 18$ ways that Alistair could have travelled to the treasure, hence (18).

28. Sami dropped his schoolbag and his ruler broke into three pieces. The two end pieces are shown.



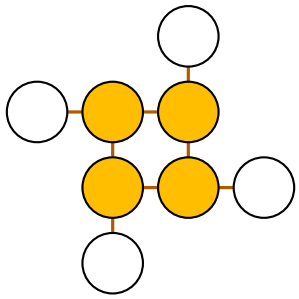
What do all the numbers on the middle piece add up to?

The numbers on the middle piece can be paired into 30s: $6 + 24$ up to $22 + 8$, with an extra 7 that isn't paired. There are 17 pairs, so that total is $17 \times 30 + 7 = 510 + 7 = 517$, hence (517).

29. The numbers 2 to 9 are to be placed in these 8 circles so that each row and each column adds to the same total.

The numbers in the 4 shaded circles are then added.

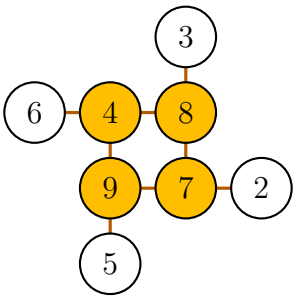
What is the largest possible total of these 4 numbers?



The total of all numbers used is $2+3+4+5+6+7+8+9 = 44 = 4 \times 11$. The total of all four indicated lines is also a multiple of 4. Consequently the 4 shaded numbers, which are counted twice in this second total, must be a multiple of 4.

The total in the 4 shaded circles is at most $6+7+8+9 = 30$, but to be a multiple of 4, 28 is the most. Then the total of the 4 lines is $28 + 44 = 72$ so that each line adds to 18.

Try $4+7+8+9 = 28$ in the 4 shaded circles, but $8+9$ can't be on a common line, so we place 8 and 9 diagonally, and then fill in the remainder using arithmetic:



Since the total is at most 28, and 28 can be found, it is the largest possible total,

hence (28).

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30. Skip counting by 5, I write a list of 400 numbers that starts 5, 10, 15 and ends 1990, 1995, 2000.

How many times do I write the digit '0'?

Every second multiple of 5 ends in 0. This gives 200 zeros in the units place.

Among the three- and four-digit numbers, there is a zero in the tens place for the numbers $100, 105, 200, 205, \ldots, 1900, 1905$ and 2000. This adds $19 \times 2 + 1 = 39$ zeros.

Finally, there are extra zeros in the hundreds place for the numbers $1000, 1005, 1010, 1015, \ldots, 1090, 1095$ and 2000. This gives another $2 \times 10 + 1 = 21$ zeros.

Therefore there are $200 + 39 + 21 = 260$ zeros in total,

hence (260).